Analysis and Design of Algorithms

NUS National University of Singapore

Algorithms

CS3230

Tutorial

Week 10

Question 1

Consider an undirected graph G where all the edge weights are distinct. Which of the following statements are true?

1. The edge with the smallest weight must be included in any minimum spanning tree.
2. The edge with the second smallest weight must be included in any minimum spanning tree.
3. The edge with the third smallest weight must be included in any minimum spanning tree.

- (1) only.
- (1) and (2) only.
- All (1), (2), and (3).

Question 1

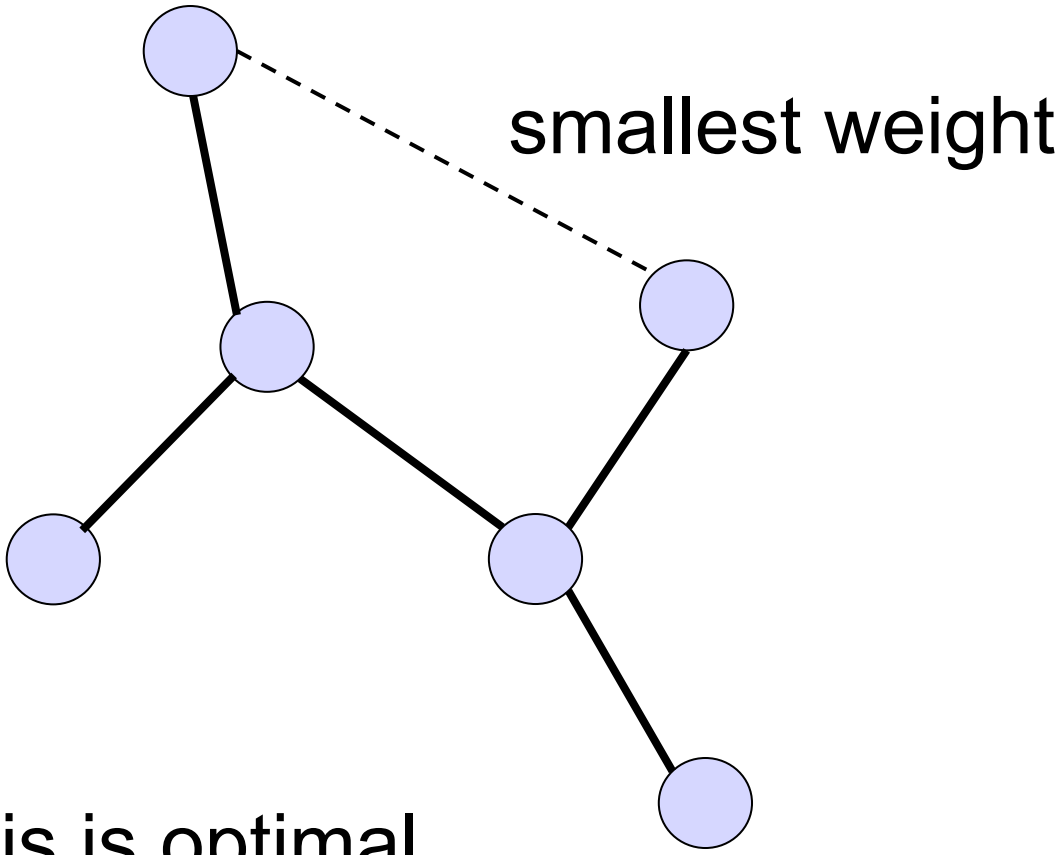
Answer: B (1) and (2) only

(I) The edge with smallest weight must be included in any minimum spanning tree

True: Consider any minimum spanning tree that does not include the smallest weight.

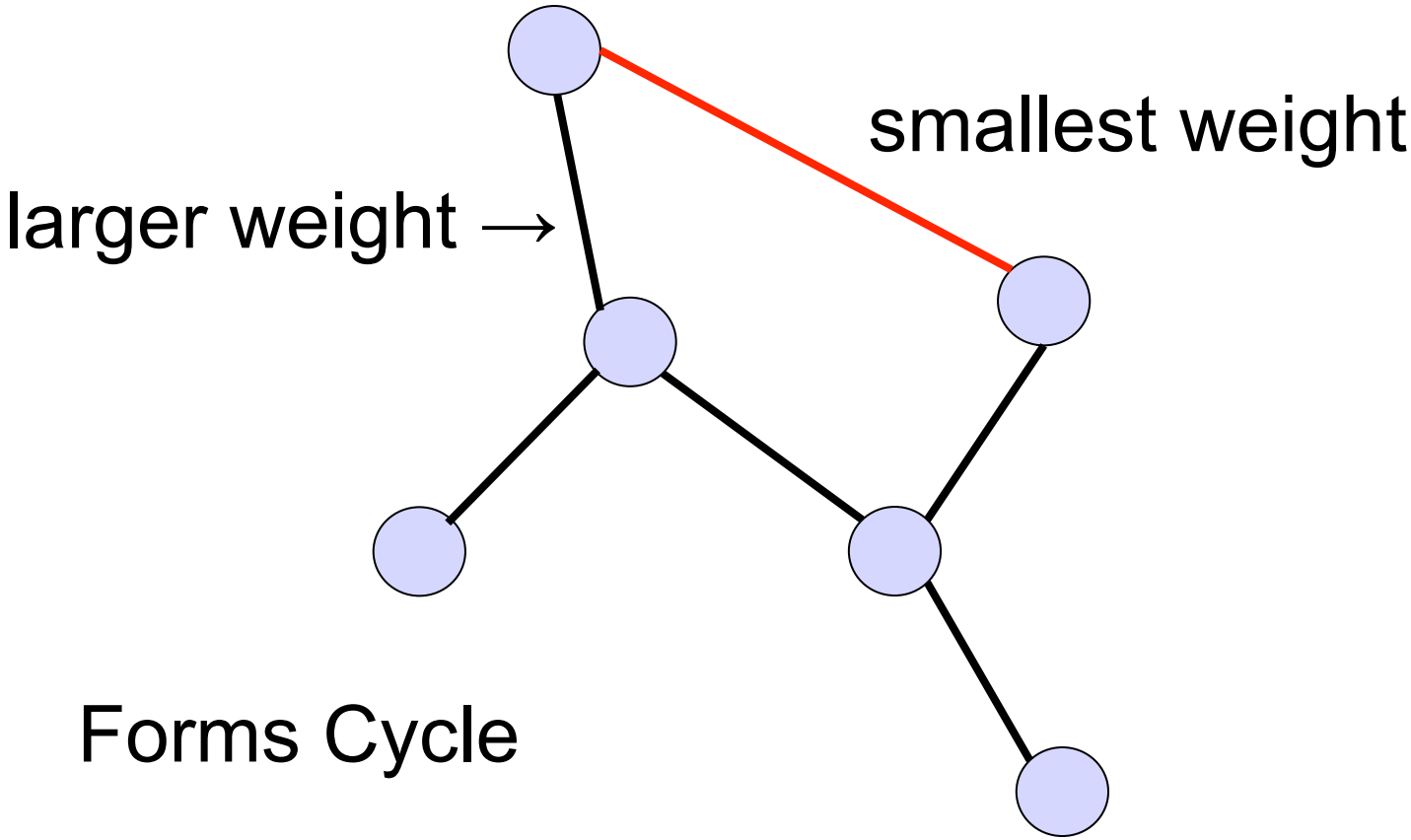
1. Add the weight in, forming a cycle.
2. Delete some other edge in the cycle, forming a tree again.
3. This tree has smaller weight, contradicting the fact that it is the spanning tree is minimum in the first place.

Question 1



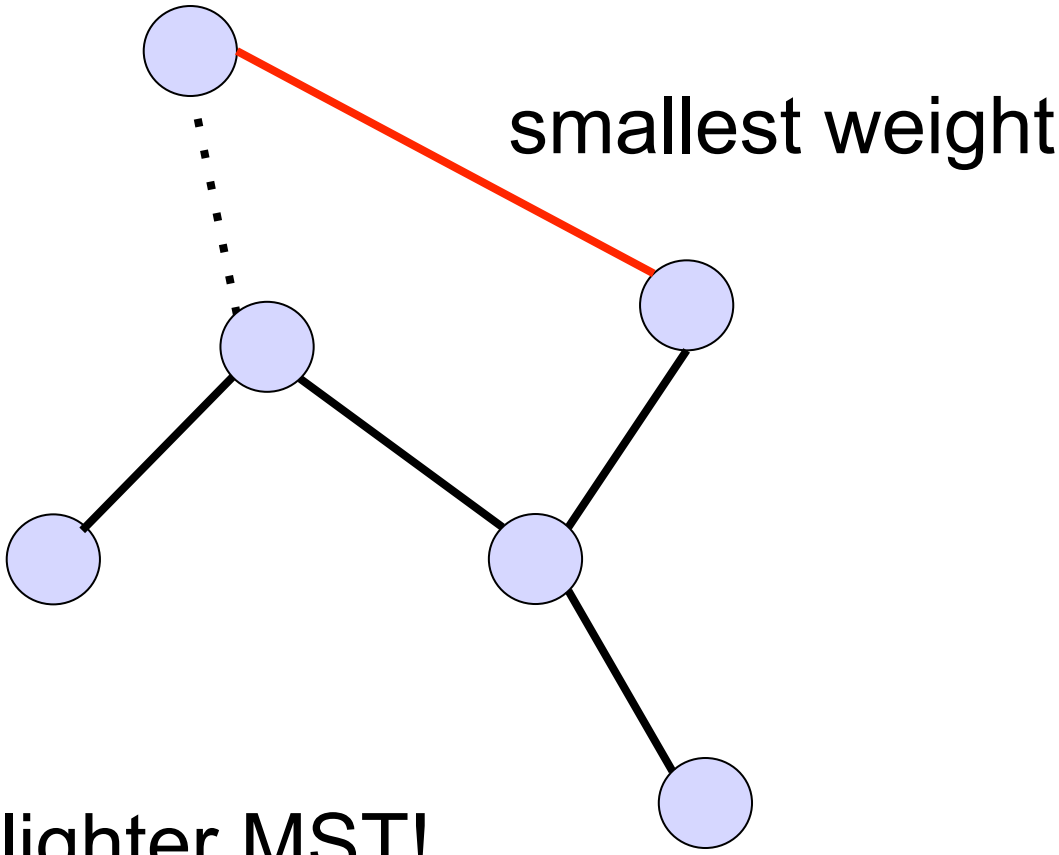
Suppose this is optimal.
And the smallest weight is not in there.

Question 1



Forms Cycle

Question 1



We have a lighter MST!

Question 1

Answer: B (1) and (2) only

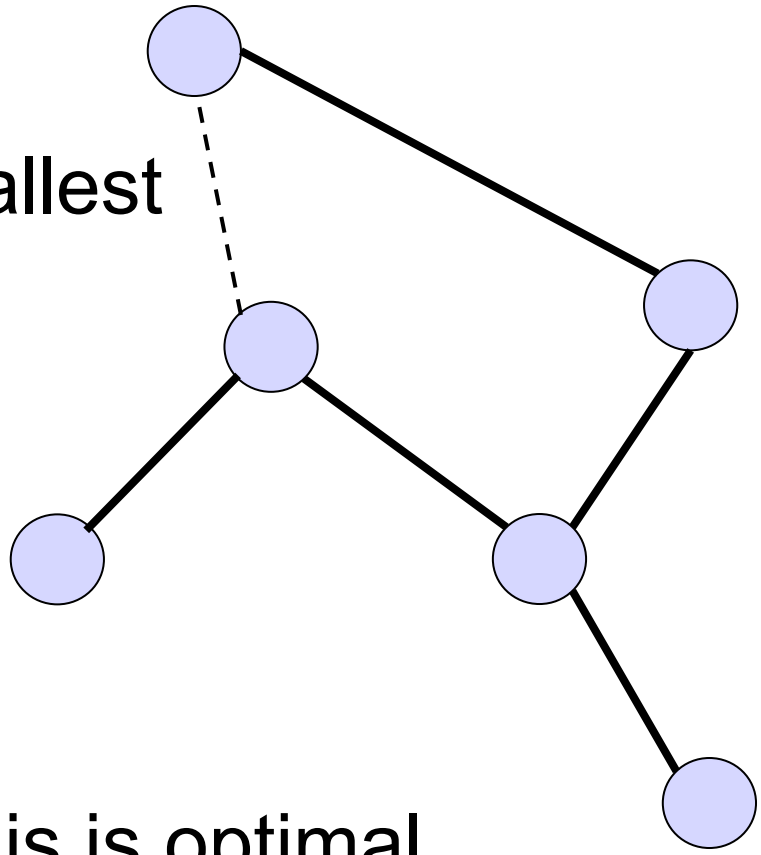
(II) The 2nd smallest weight edge must be included in any minimum spanning tree

True: Consider any minimum spanning tree that does not include the 2nd smallest weight.

1. Add the weight in, forming a cycle.
2. Delete any other edge except the two smallest, forming a tree again -- always possible as cycle has at least 2 edges
3. This tree has smaller weight, contradicting the optimality of the first tree.

Question 1

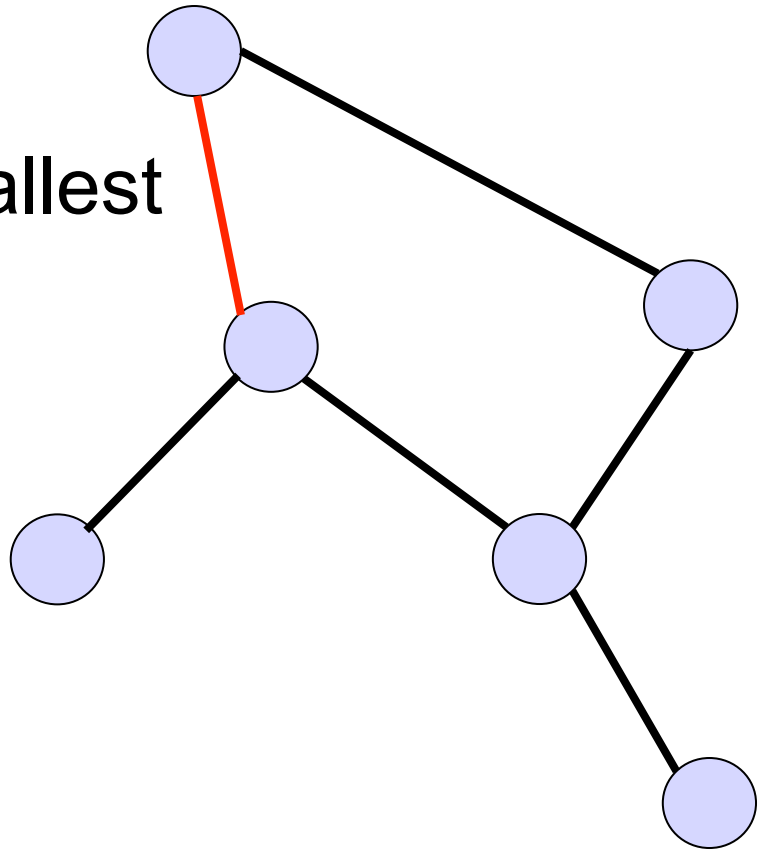
second smallest weight



Suppose this is optimal.
And the second smallest weight is not in there.

Question 1

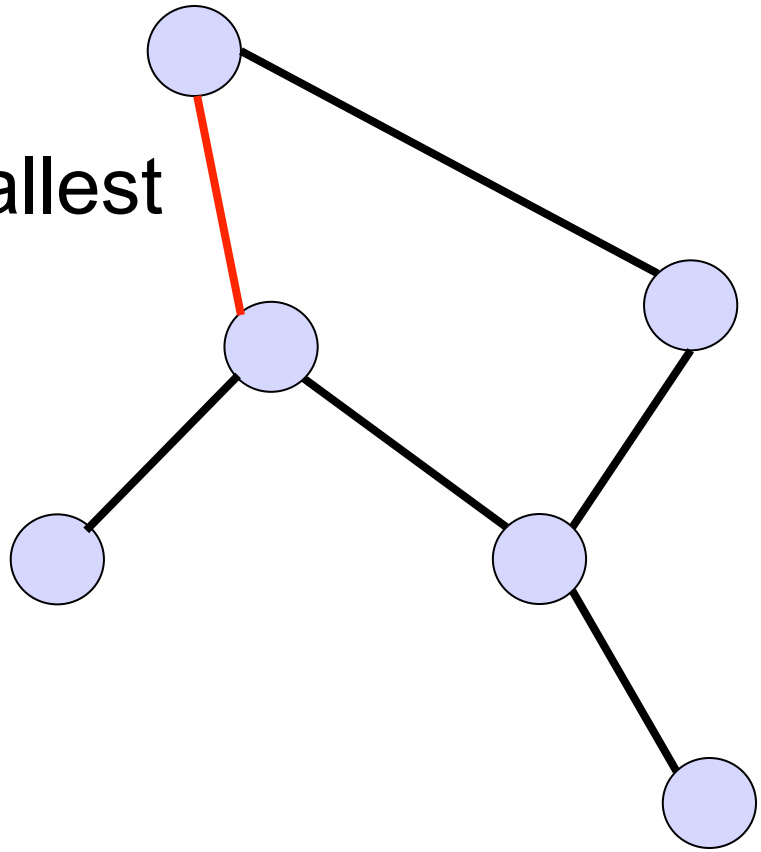
second smallest weight



Add it in, and you form a cycle.

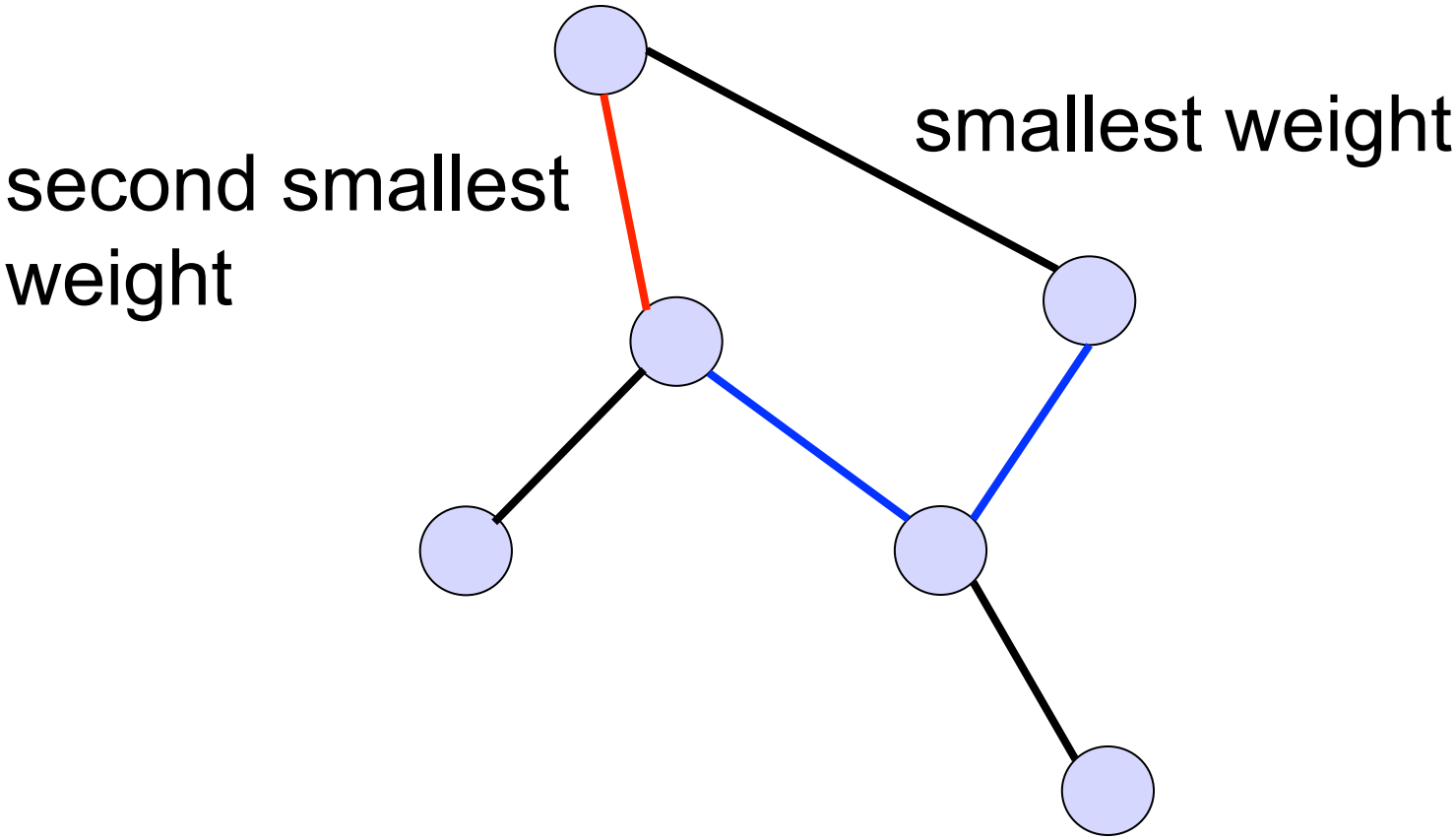
Question 1

second smallest weight



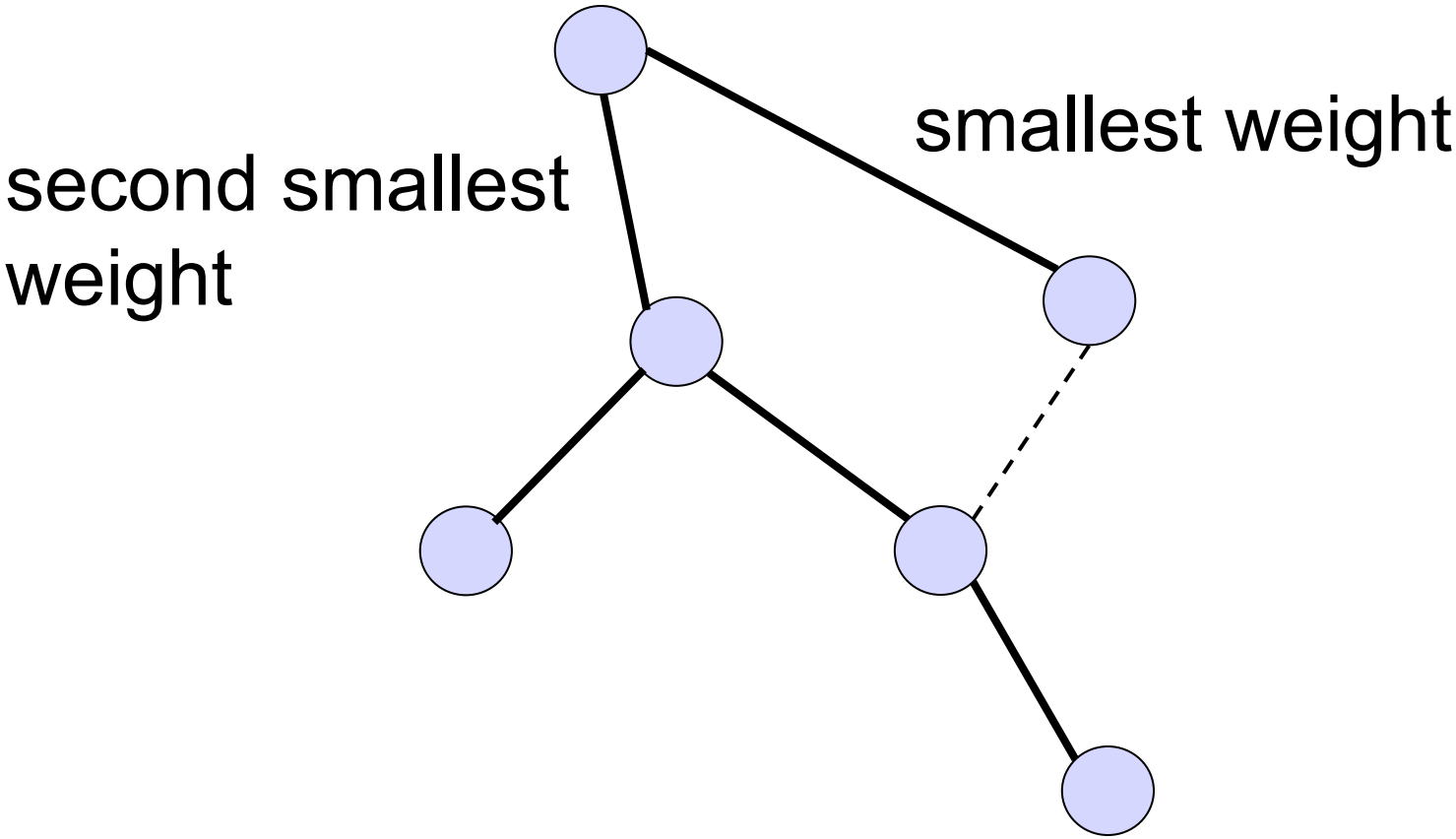
The cycle must have at least **3** edges.
(else it wouldn't be a cycle)

Question 1



One of them must not be the smallest weight edge.

Question 1



Remove an edge that must not be the smallest weight, and you have a lighter tree!

Question 1

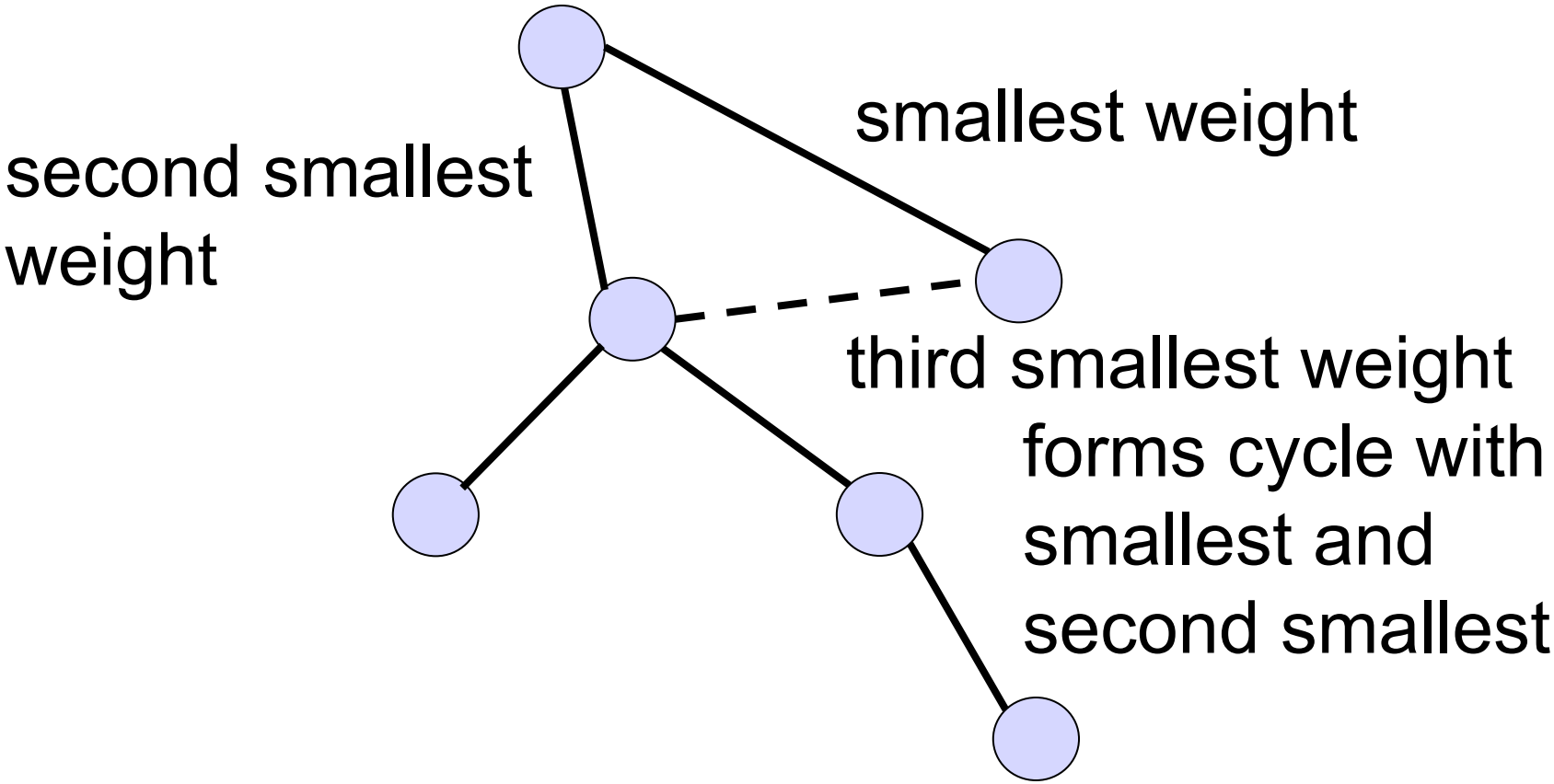
Answer: B (1) and (2) only

(II) The 3rd smallest weight edge must be included in any minimum spanning tree

False: Consider a graph with three vertices and three edges.

1. The two smallest weights form the spanning tree.
2. Adding the third will form a cycle.

Question 1



The third smallest edge need not be in an MST!

Question 2

Suppose Bob has a collection of music files that he wants to burn into CDs. Each CD has a storage capacity of 100MB. In addition, Bob does not want to store more than two music files per CD. Note that each music file cannot be split and hence cannot be burned on more than 1 CDs. Given a set A of file sizes, each smaller than 100MB, let $MinCD(A)$ denote the minimum number of CDs required to fit the files described in A.

Which of the following correctly describes an optimal substructure property of the problem, assuming that at least one pair of files fit onto a CD?

- For any pair of files f_1, f_2 in A$MinCD(A) = 1 + MinCD(A \setminus \{f_1, f_2\})$.

- For any pair of files f_1, f_2 in A that belong on a single CD in an optimal solution, $MinCD(A) = 1 + MinCD(A \setminus \{f_1, f_2\})$.

- If f_1 and f_2 are the largest and smallest files in A,$MinCD(A) = 1 + MinCD(A \setminus \{f_1, f_2\})$.

Question 2

Answer: B

Answer A does not work as we cannot simply pair up any two file and still be optimal.

Counterexample to A:

Consider {10, 20, 80, 90}. Optimal solution is {10,90}, {20, 80}.

MinCD({10,20,80,90}) ≠ 1 + *MinCD*({80,90})

after pairing {10,20} as {80,90} requires 2 CDs

Question 2

Answer: B

Answer C does not work as the largest and smallest file may not fit into a single CD.

Counterexample to A:

Consider {10, 20, 95}. Optimal solution is {10,20}, {95}.

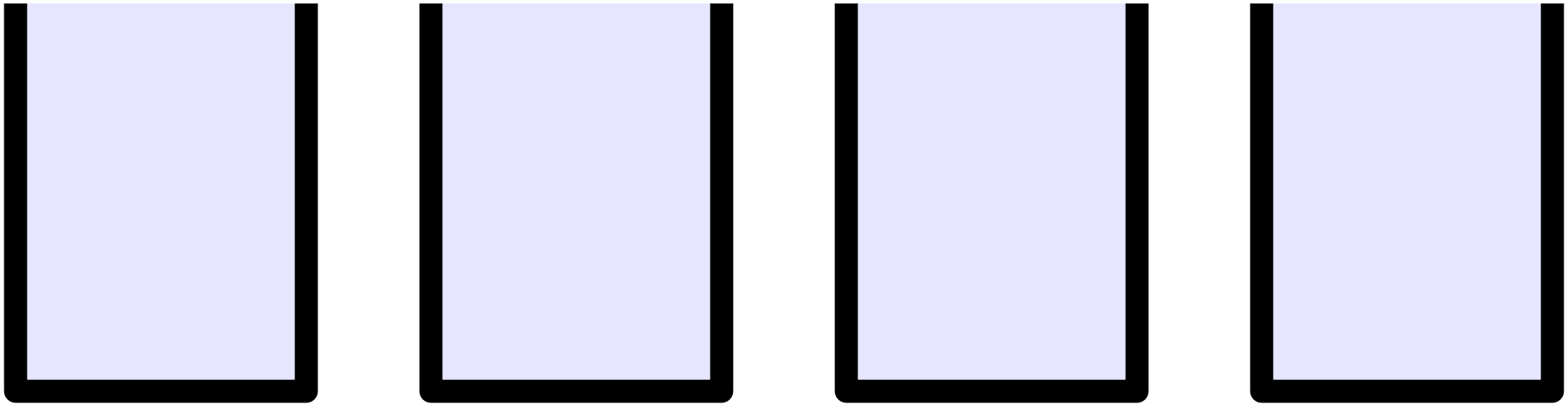
If we remove 10 and 95, need two CDs to store them. Then left with 20 which requires another CD.

Question 2

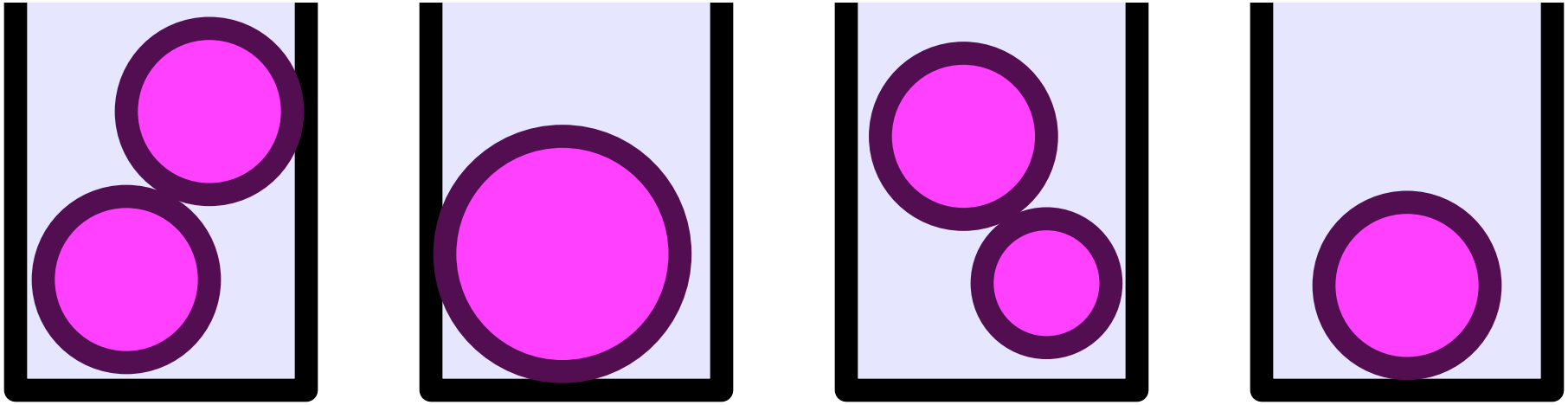
Answer: B

Answer B is correct. Consider any optimal solution.

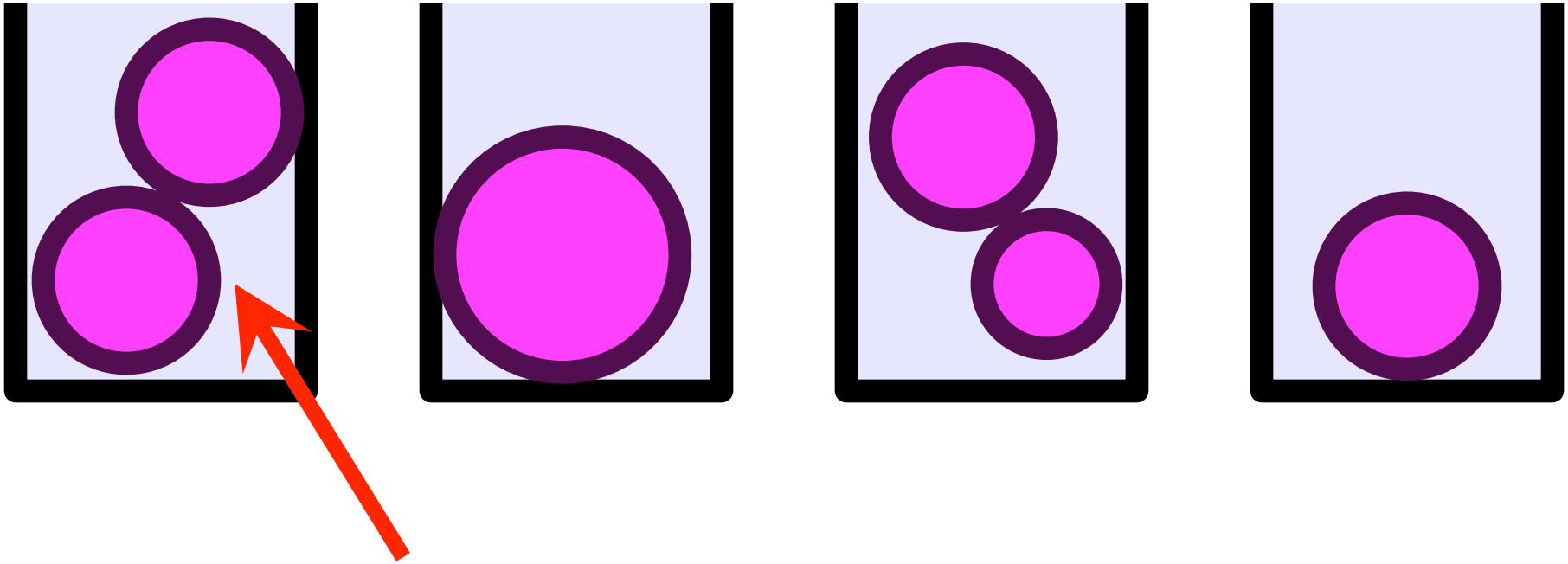
1. We can remove any pair that is in the same CD in the optimal solution.
2. The rest of the files must be stored optimally, otherwise can reduce the total number of CDs used (**optimal substructure**).



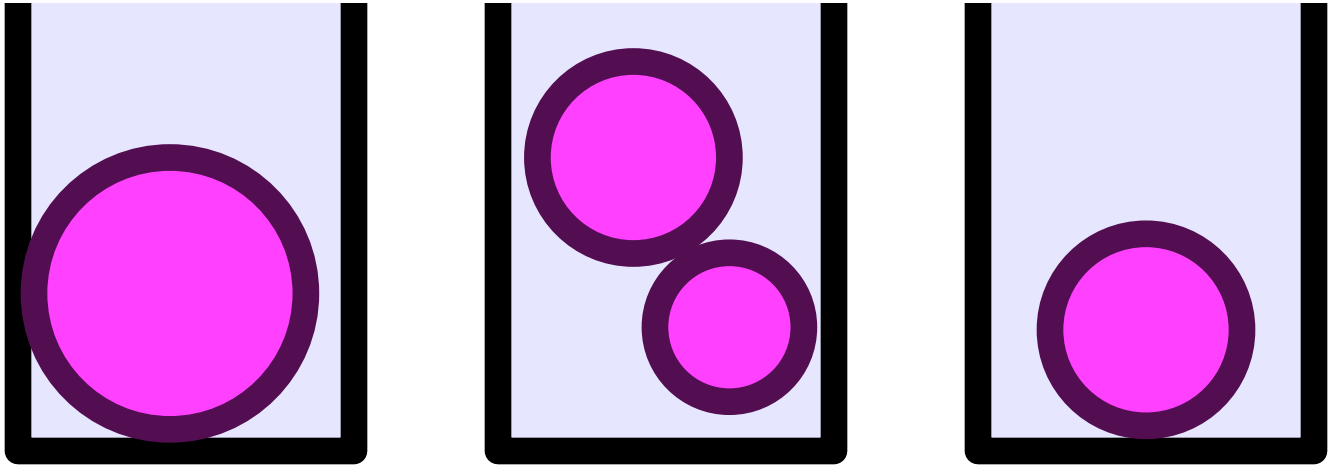
Optimal Solution:



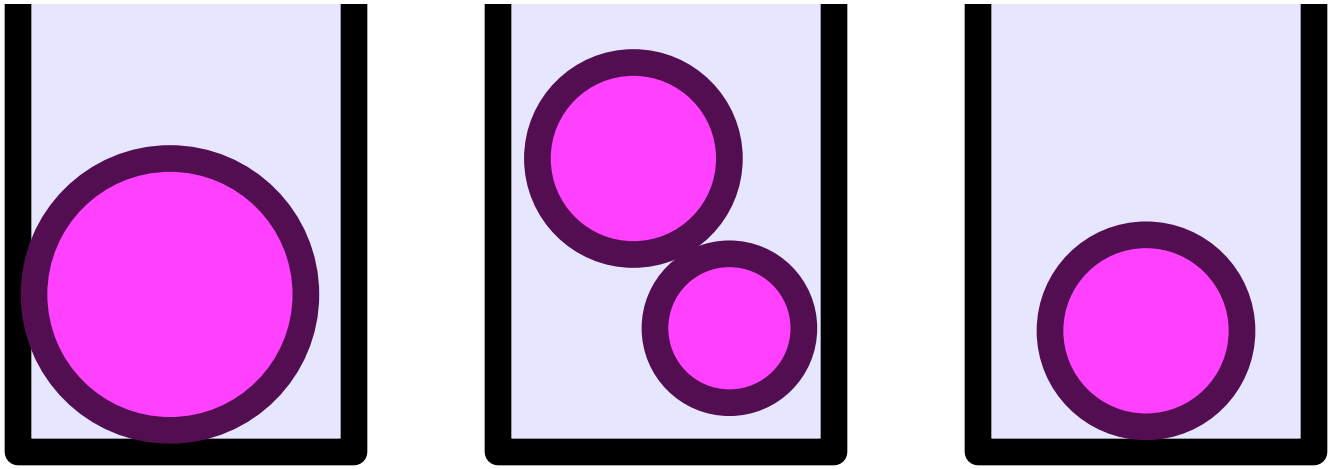
Optimal Solution:



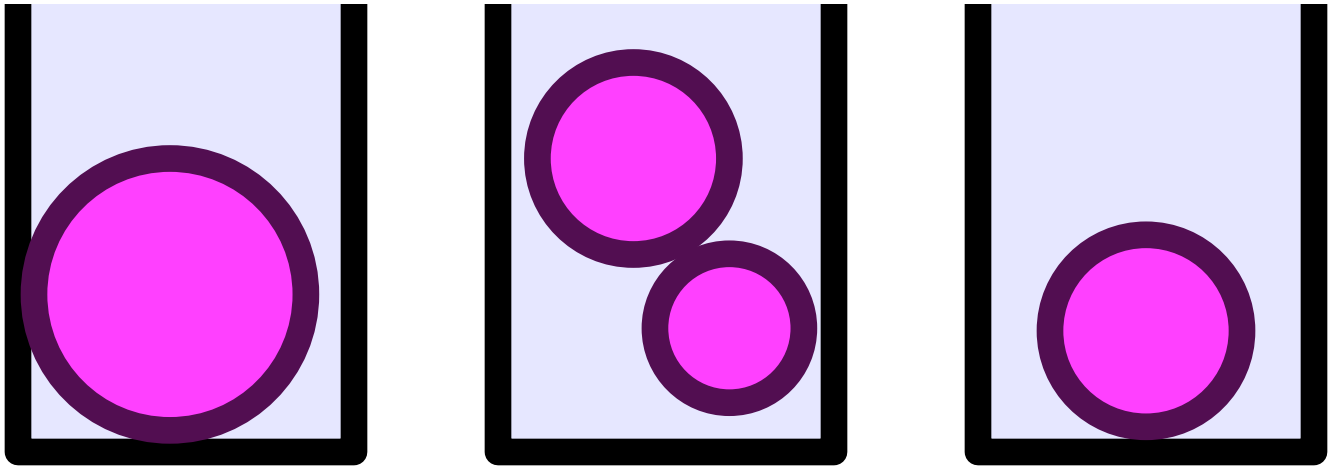
These two are paired together
in an optimal solution



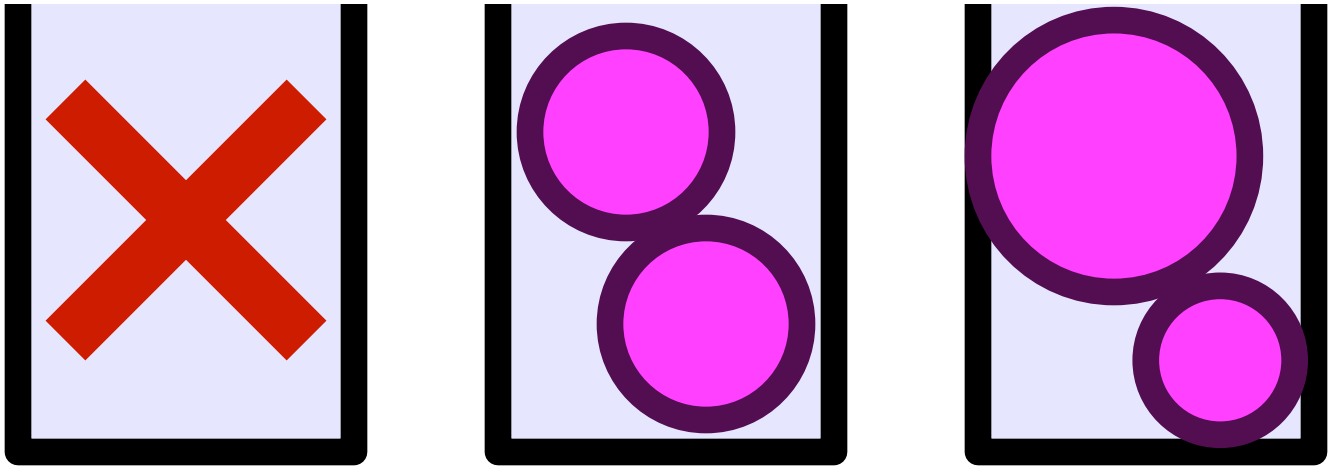
This must be optimal too!



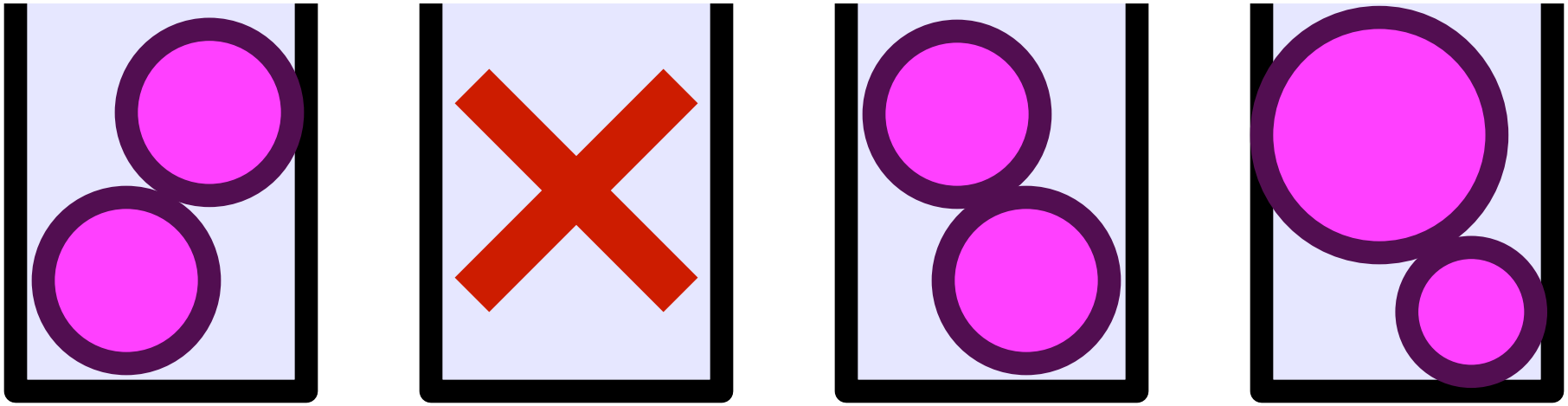
Suppose Not.



Then we can find some allocation with less CDs



Then the original is not optimal!



Question 3

Suppose Bob has a collection of music files that he wants to burn into CDs. Each CD has a storage capacity of 100MB. In addition, Bob does not want to store more than two music files per CD. Note that each music file cannot be split and hence cannot be burned on more than 1 CDs. Given a set A of file sizes, each smaller than 100MB, let $MinCD(A)$ denote the minimum number of CDs required to fit the files described in A.

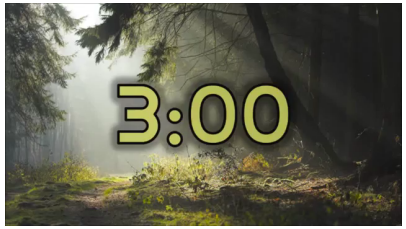
Assume that any optimal solution contains a pair that is burned onto a CD. Select all true statements about the problem.

(I) The smallest file must be included in a pair in some optimal solution.

(II) The pair $\{f_1, f_2\}$ where f_1 is the smallest file and f_2 is the largest file such that f_1 and f_2 fit onto one CD must be included in a pair in some optimal solution.

(III) The pair $\{f_1, f_2\}$ where f_1 is the smallest file and f_2 is the largest file must be included in a pair in some optimal solution.

- (I) only
- (I) and (II) only
- (I), (II), and (III)



Question 3

Answer: B (I) and (II) only

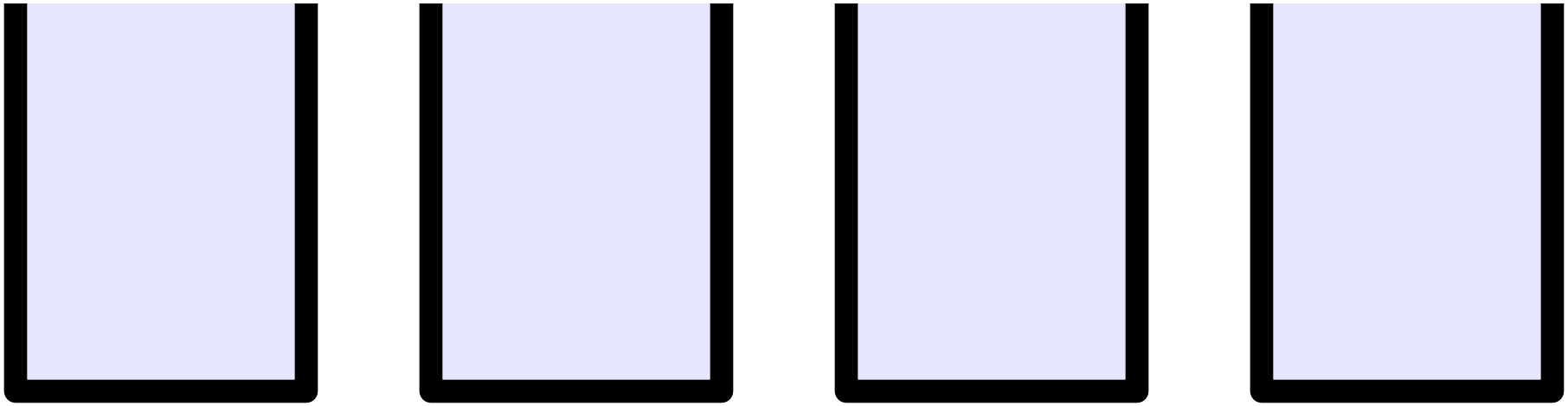
(I) If smallest file is not included in a pair, swap it with any file that is included in a pair in an optimal solution and the number of CDs used will not change.
Hence, there exists an optimal solution that contains the smallest file in a pair, if an optimal solution with a pair exists. (**Greedy choice**)

Question 3

Answer: B (I) and (II) only

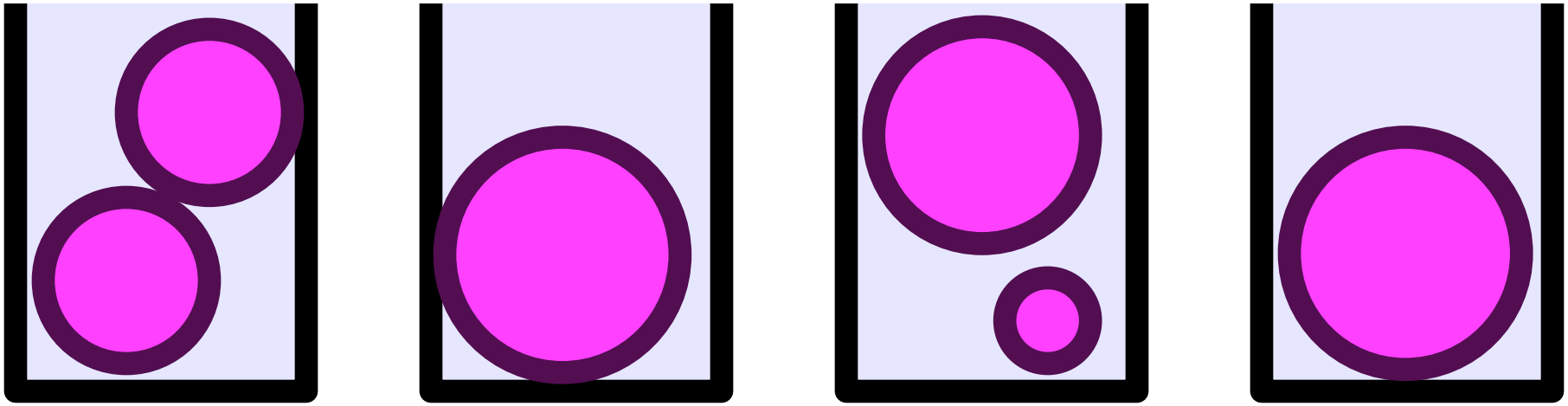
(II) From part (I), we know there is an optimal solution where smallest file is in a pair, if optimal solution contains pair.

Assume we have optimal solution where smallest file is paired with F but the largest file that fits is F'. Swap F with F'. If F' originally not paired, we still have an optimal solution. If F' paired, swap still possible as F' larger, hence F must fit in swapped CD. Hence, optimal solution with F' paired with smallest file exists. (**Greedy choice**)



(I)

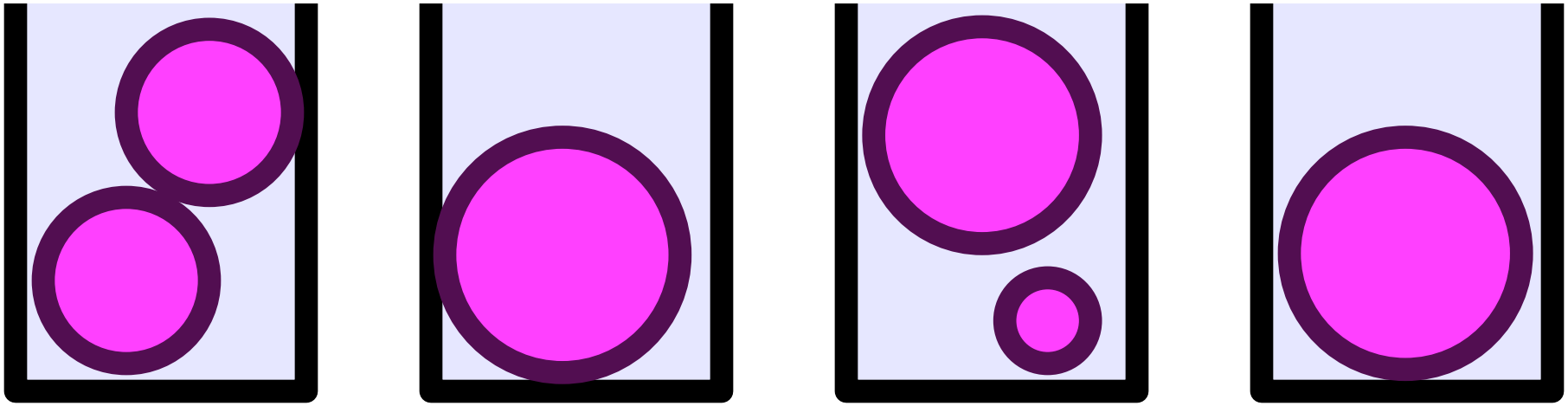
Take any Optimal Solution



[Optimal]

(I)

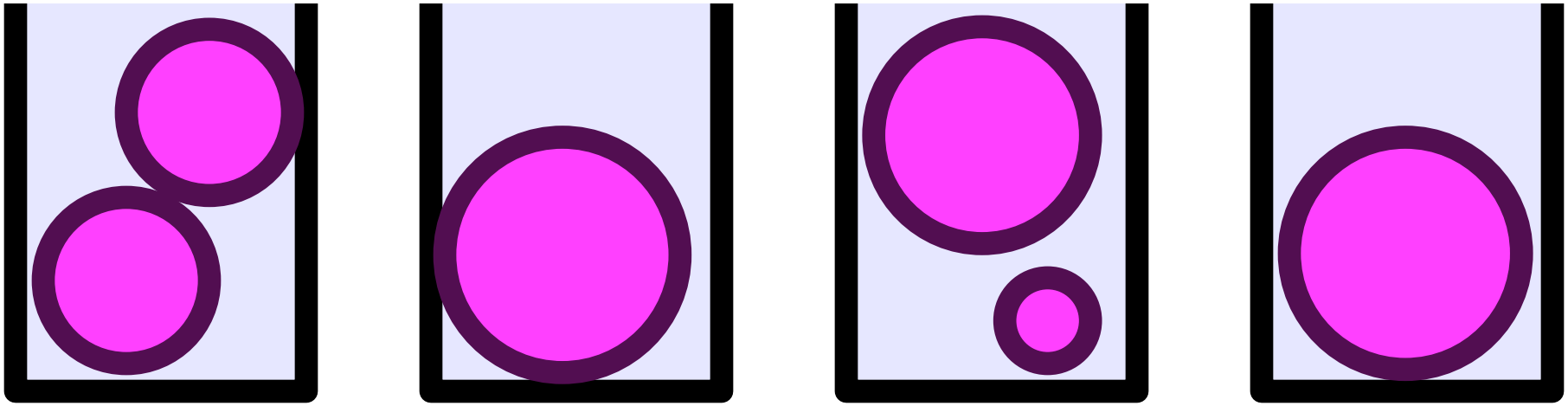
Either the smallest is in a pair,
Or it is not.



[Optimal]

(I)

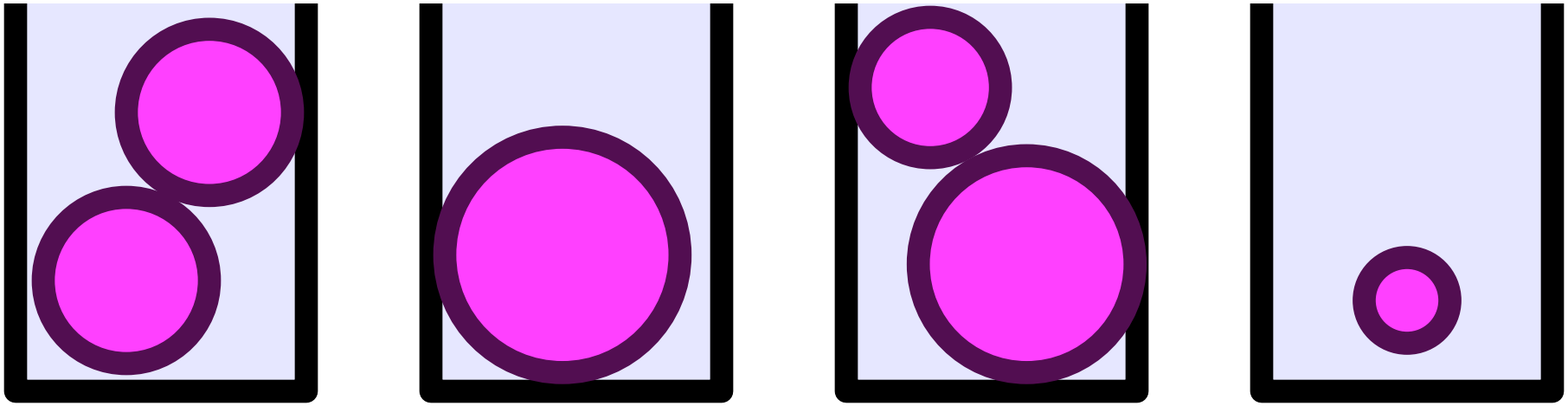
If it is, we are done.



[Optimal]

(I)

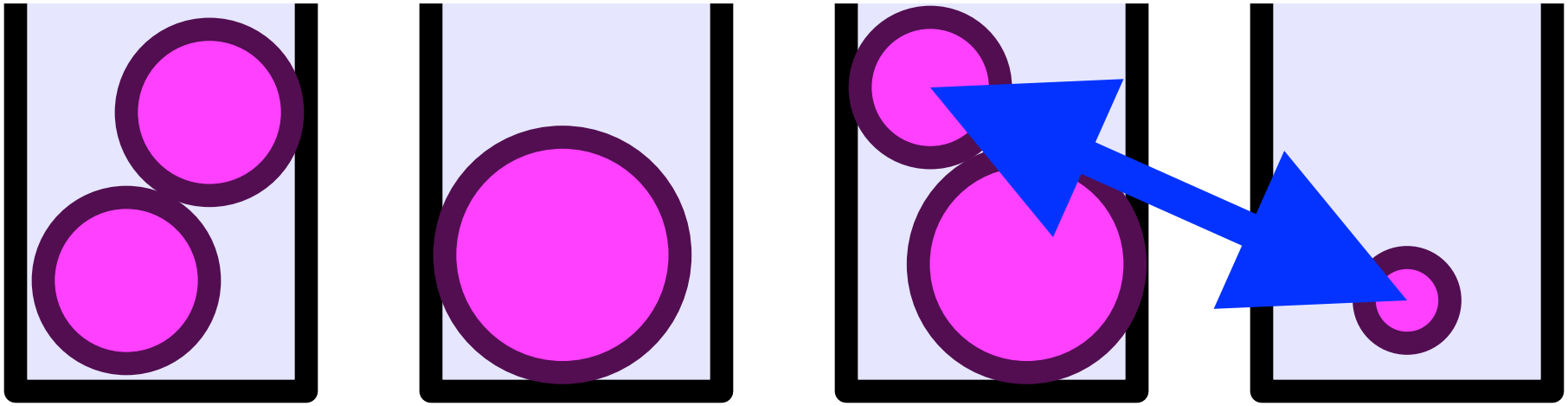
If it is not,



[Optimal]

(I)

We can always swap.

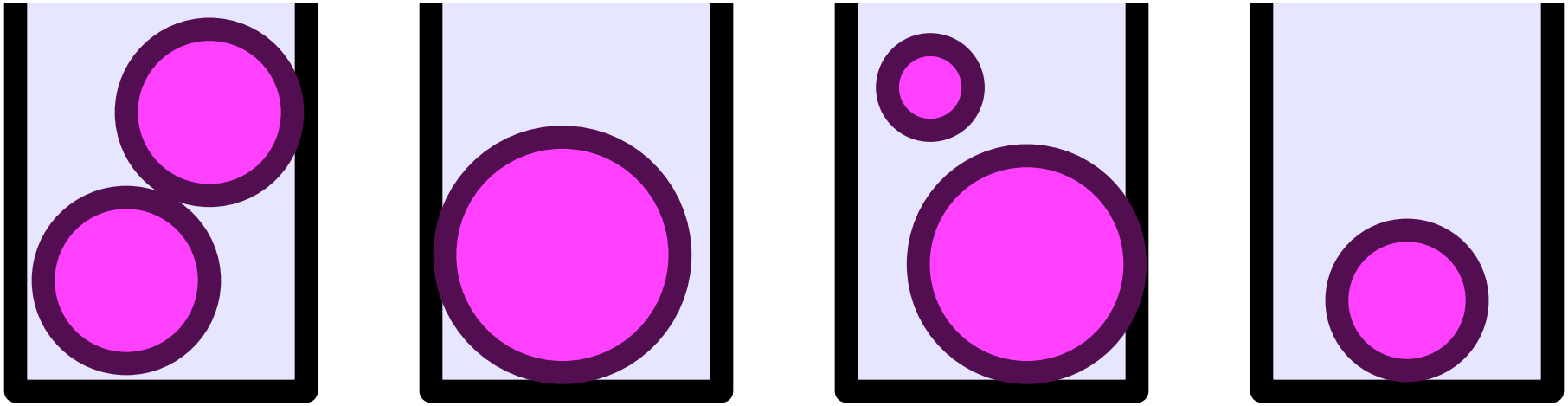


[Optimal]

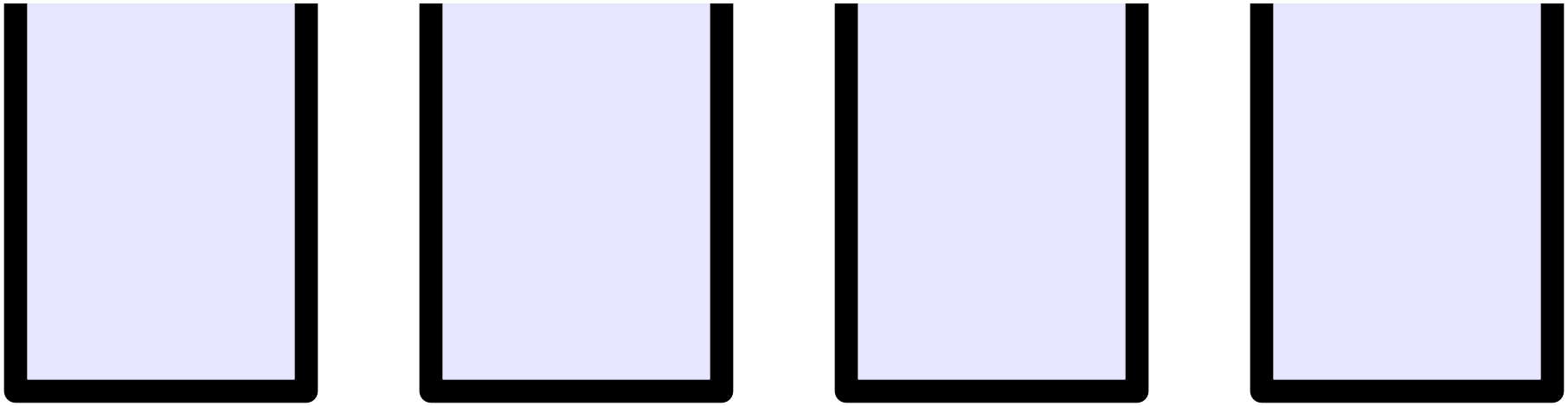
(I)

This is still optimal.

There is indeed an optimal solution where the smallest element is paired.

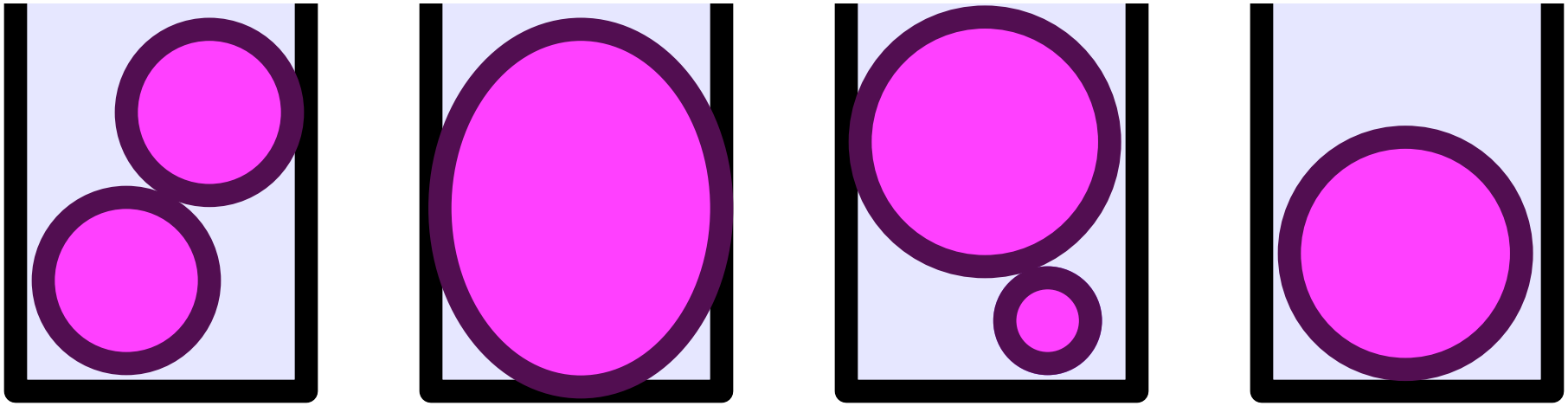


[Optimal]



(II)

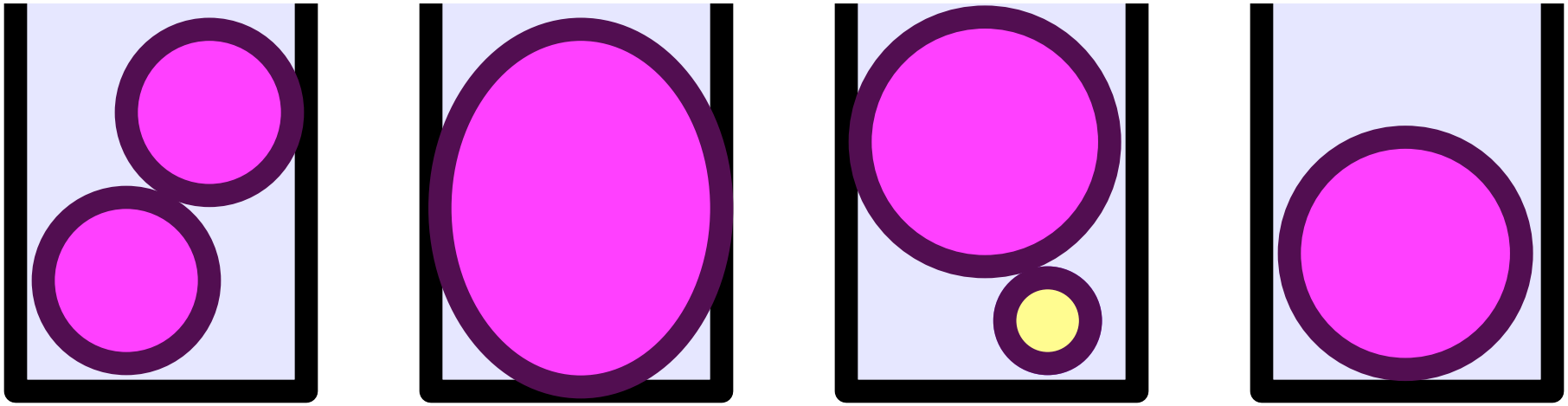
Take any Optimal Solution



[Optimal]

(II)

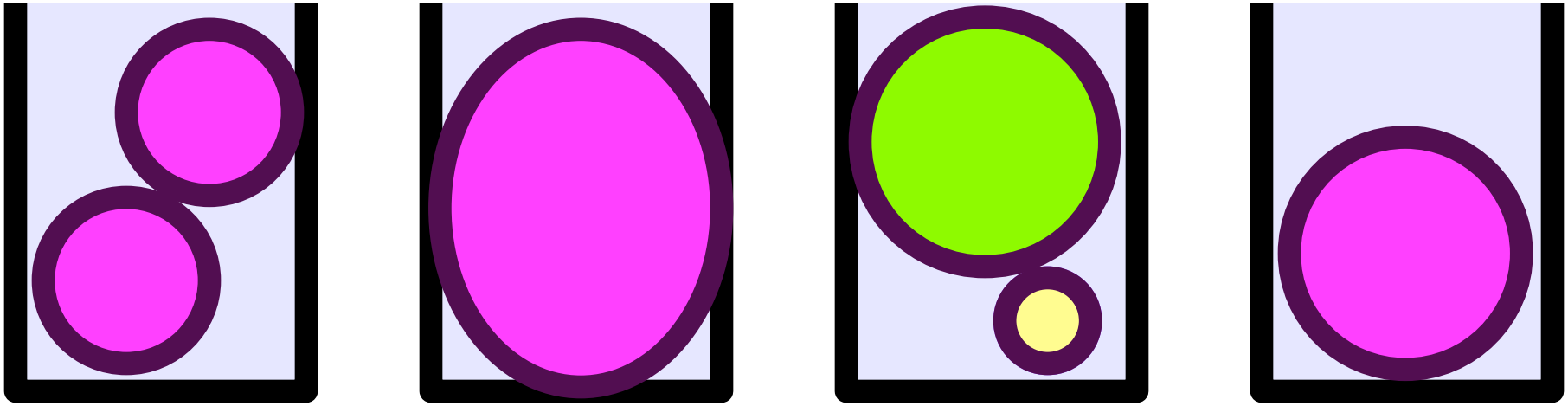
Look at the smallest element.



[Optimal]

(II)

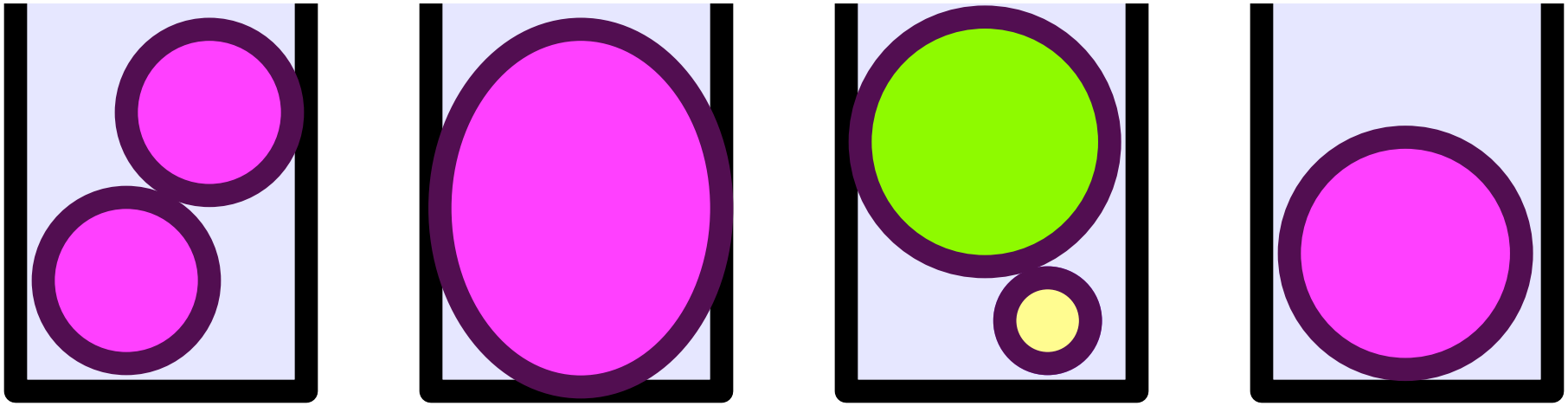
Look at the largest element That can fit with the smallest



[Optimal]

(II)

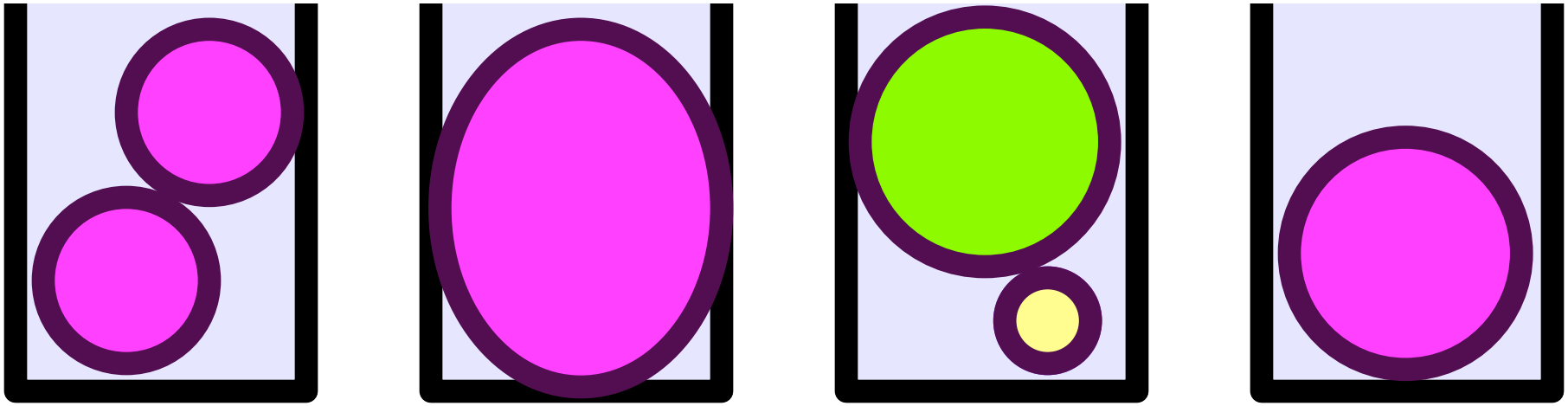
They are either in the same bin,
Or they are not.



[Optimal]

(II)

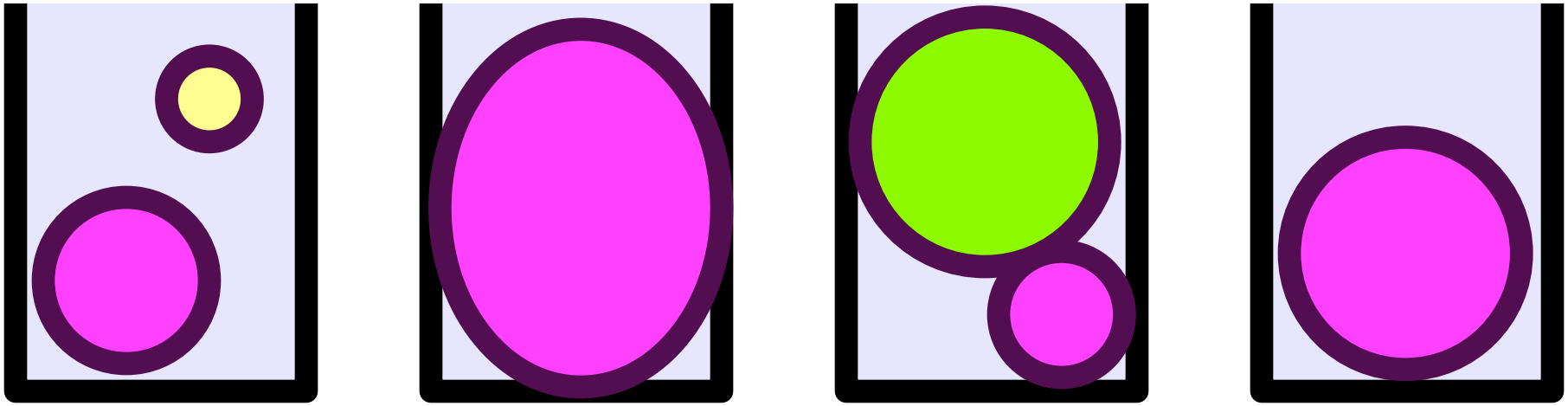
If they are, we are done.



[Optimal]

(II)

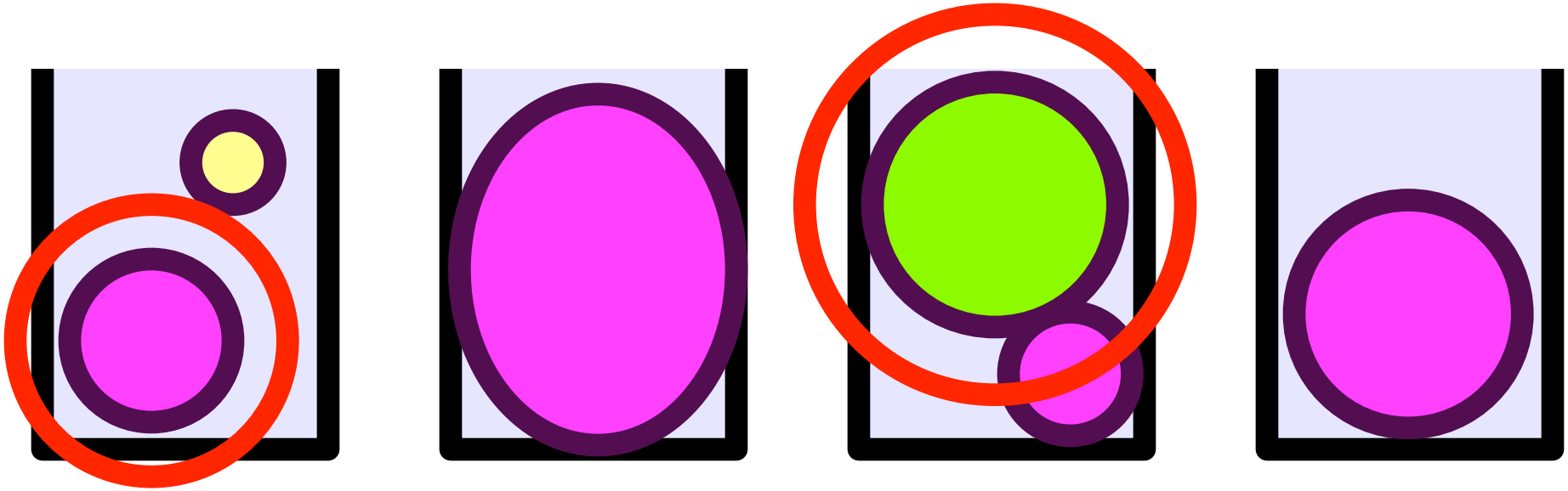
If they are not,



[Optimal]

(II)

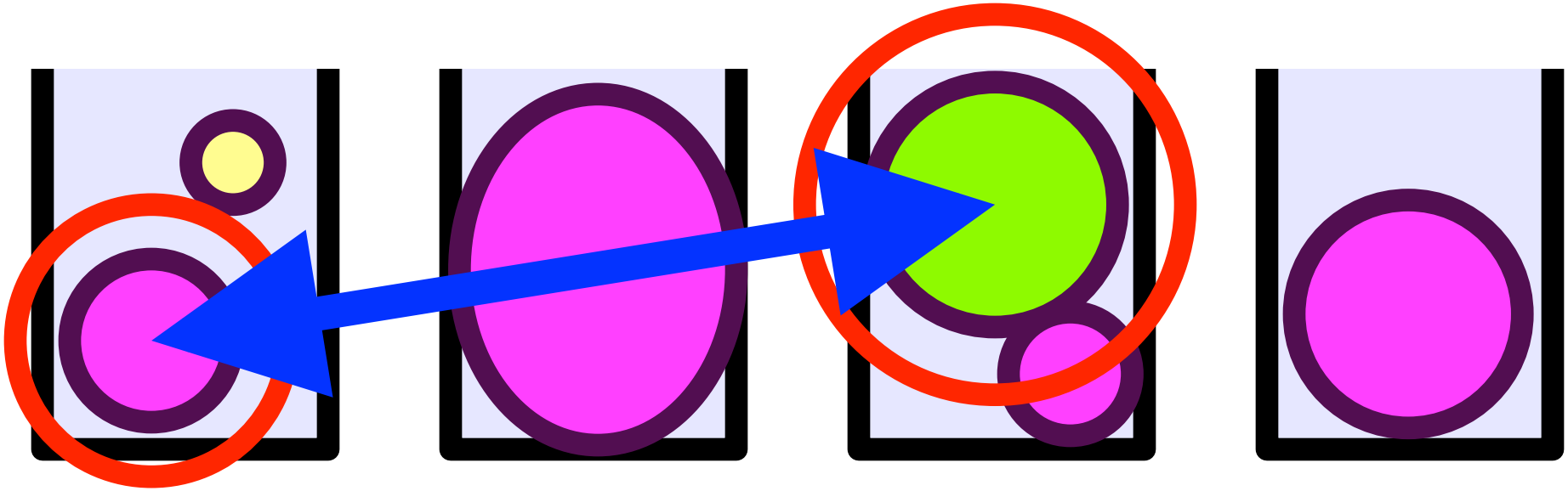
Look at these two elements.



[Optimal]

(II)

The smallest element's current partner
Can fit with the (largest element that fits with the smallest element)'s current partner.

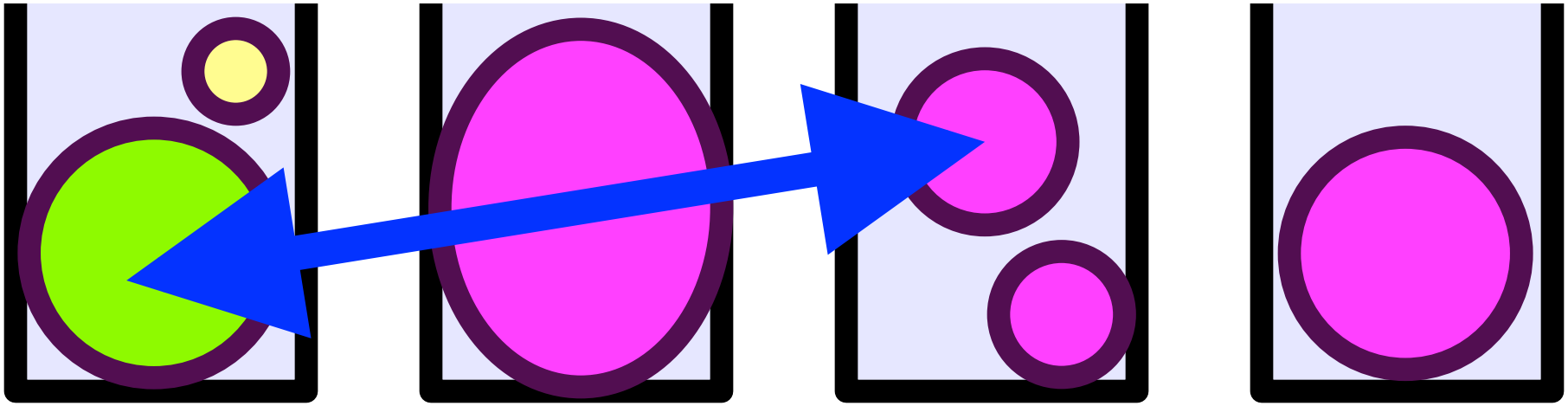


[Optimal]

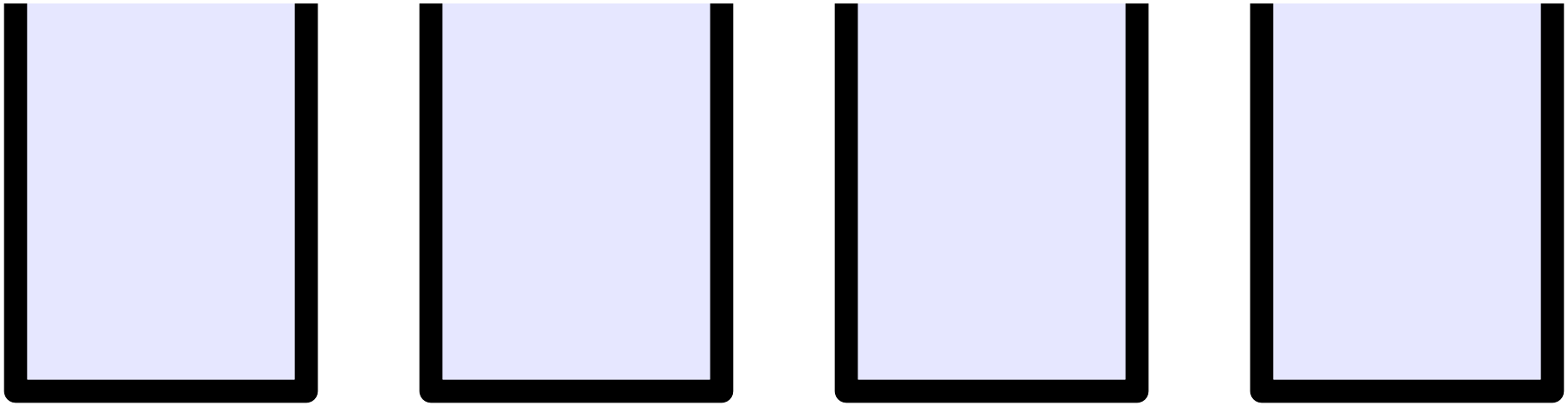
The largest element that fits with the smallest element
Can fit with the smallest element (duh)

(II)

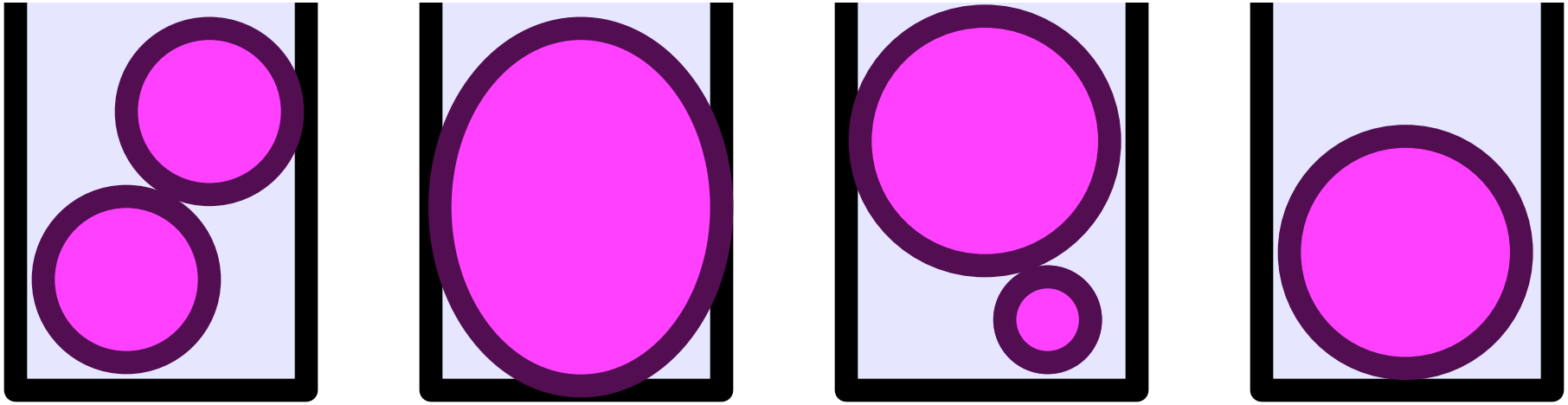
No new CDs created.
Still optimal.



[Optimal]

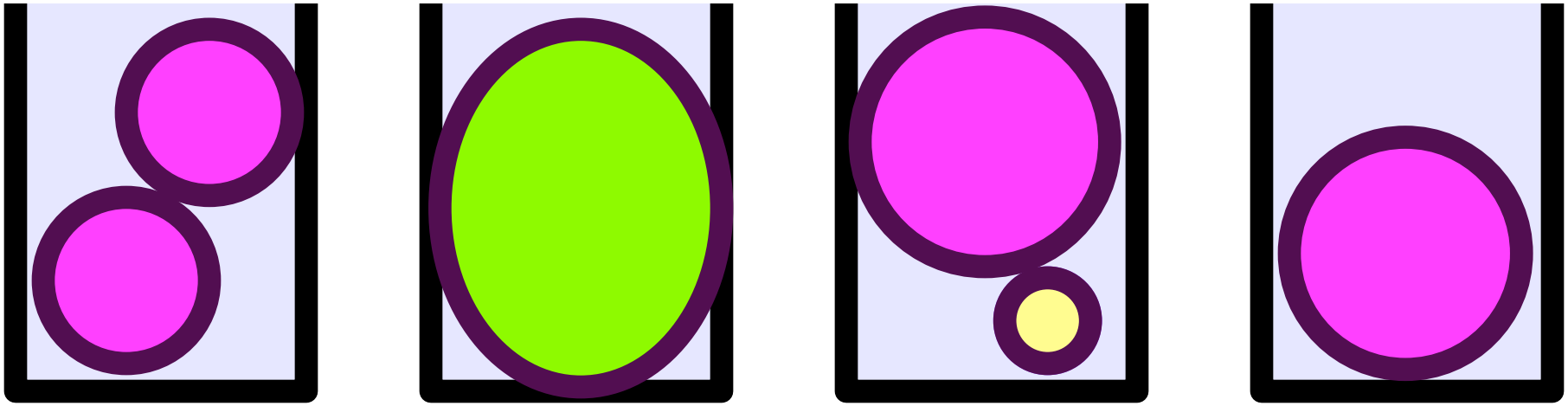


(III) NOT TRUE



(III) NOT TRUE

The smallest and largest elements May not fit together at all!



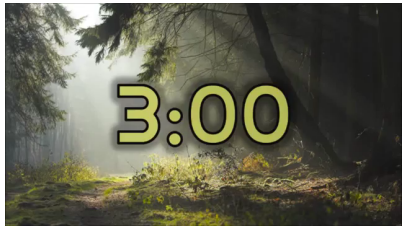
Question 4

Suppose Bob has a collection of music files that he wants to burn into CDs. Each CD has a storage capacity of 100MB. In addition, Bob does not want to store more than two music files per CD. Note that each music file cannot be split and hence cannot be burned on more than 1 CDs. Derive the algorithm for obtaining the the minimum number of CDs that Bob needs to burn his music files on the discs and apply it to the following array of file sizes.

$filesizes = [89, 59, 32, 74, 81, 12, 7, 49, 43, 51, 62, 91, 27]$

What is the minimum number of CDs required?

- 8
- 7
- 10
- 9



Question 4

Answer: A. 8

[89, 59, 32, 74, 81, 12, 7, 49, 43, 51, 62, 91, 27]

Sort the array

[7, 12, 27, 32, 43, 49, 51, 59, 62, 74, 81, 89, 91]

Use two pointers, one moving forward from start, another moving backwards from end.

Runtime $O(n \log n)$ from sorting.

- Run the greedy algorithm to get {7,91}, {12,81}, {27, 62}, {32, 59}, {43, 51}, {49}, {74}, {89}

Question 4

Answer: A. 8

[**7**, 12, 27, 32, 43, 49, 51, 59, 62, 74, 81, 89, 91]

- Run the greedy algorithm to get {**7**,91}, {12,81}, {27, 62}, {32, 59}, {43, 51}, {49}, {74}, {89}

Question 4

Answer: A. 8

[**7**, 12, 27, 32, 43, 49, 51, 59, 62, 74, 81, 89, **91**]

- Run the greedy algorithm to get **{7,91}**, {12,81}, {27, 62}, {32, 59}, {43, 51}, {49}, {74}, {89}

Question 4

Answer: A. 8

[**7**, **12**, 27, 32, 43, 49, 51, 59, 62, 74, 81, 89, **91**]

- Run the greedy algorithm to get **{7,91}**, {**12**,81}, {27, 62}, {32, 59}, {43, 51}, {49}, {74}, {89}

Question 4

Answer: A. 8

[**7**, **12**, 27, 32, 43, 49, 51, 59, 62, 74, **81**, 89, **91**]

- Run the greedy algorithm to get **{7,91}**, **{12,81}**, {27, 62}, {32, 59}, {43, 51}, {49}, {74}, {89}

Question 4

Answer: A. 8

[**7**, **12**, **27**, 32, 43, 49, 51, 59, 62, 74, **81**, 89, **91**]

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Question 4

Answer: A. 8

[**7**, **12**, **27**, 32, 43, 49, 51, 59, **62**, 74, **81**, 89, **91**]

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[**7**, **12**, **27**, **32**, 43, 49, 51, 59, **62**, 74, **81**, 89, **91**]

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[**7**, **12**, **27**, **32**, 43, 49, 51, **59**, **62**, 74, **81**, 89, **91**]

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Question 4

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[**7**, **12**, **27**, **32**, **43**, 49, 51, **59**, **62**, 74, **81**, 89, **91**]

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Question 4

Answer: A. 8

[**7**, **12**, **27**, **32**, **43**, 49, **51**, **59**, **62**, 74, **81**, 89, **91**]

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[**7**, **12**, **27**, **32**, **43**, **49**, **51**, **59**, **62**, **74**, **81**, 89, **91**]

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Question 4

Answer: A. 8

[**7**, **12**, **27**, **32**, **43**, **49**, **51**, **59**, **62**, **74**, **81**, **89**, **91**]

- Run the greedy algorithm to get **{7,91}**, **{12,81}**, **{27, 62}**, **{32, 59}**, **{43, 51}**, **{49}**, **{74}**, **{89}**

Question 4

Answer: A. 8

[**7**, **12**, **27**, **32**, **43**, **49**, **51**, **59**, **62**, **74**, **81**, **89**, **91**]

- Run the greedy algorithm to get **{7,91}**, **{12,81}**, **{27, 62}**, **{32, 59}**, **{43, 51}**, **{49}**, **{74}**, **{89}**
- **Total 8 CDs required!**